A Bibliometric Corpus on Central Bank Digital Currencies

Data, methods and first results of a paper in progress

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M2 – Private crypto

The paper in one slide

Research question. Why do researchers, central bankers and journalists write about CBDCs, and how has the agenda shifted since 2014?

What is written? Who writes? Why (stated motivations)? How (tone)? When (around which events)?

Contribution.

1. Data: research, speeches, official documents and press, in 45 languages: 17 times the largest earlier review.
2. Measurement: relevance filter, motivations and tone, each checked against hand coding.
3. Dynamics: attention tested against policy events with placebo inference.

48,188 collected $\rightarrow$ 17,709 with CBDC as a theme $\cdot$ 2014–2026

Related work: 77 reviews already exist

Five recent bibliometric reviews:

Study	Data	Documents	Tools
Kvedaravičiūtė & Šapkauskienė (2024)	WoS + Scopus	228	VOSviewer
Fahad & Bulut (2024)	Scopus	169 of 323	Biblioshiny
Bapat & Sharma (2025)	Scopus	371	bibliometrics + TCCM
Prodan et al. (2024)	Scopus + WoS	747 of 1,024	PRISMA, VOSviewer
Paulovici (2026)	Scopus	892	VOSviewer
This study	**5 databases, speeches, press...**	**17,709**	**+ text as data**

- Gap: one or two databases (Scopus, WoS), mostly English journal articles: central bankers and the press are absent.
- Maps say *what* is studied, not *why*; no measure is validated.

Found in our own corpus: 77 titles announce a literature, systematic or bibliometric review of CBDCs (2019–2026).

Outline

Collecting the texts

Scoring relevance

The corpus

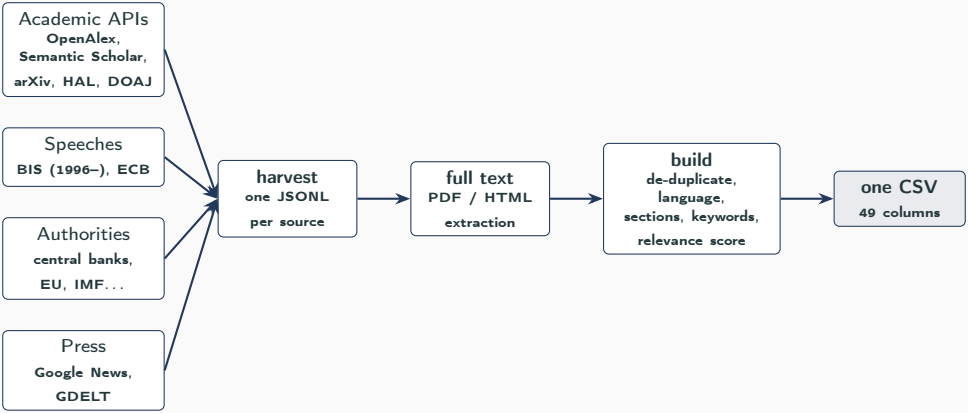
Mapping the field

Text as data: what, why, how, when

Conclusion

Collecting the texts

Pipeline



- 115 search phrases, 12 languages: *CBDC, digital euro, e-CNY, eNaira, Drex, digital fiat currency, Fedcoin, euro numérique, digitaler Euro, digital ruble...*
- Recall first: nothing is filtered at harvest time; every step is resumable.

Press: recursive bisection of time windows

```
stack = monthly_windows("2000-01-01", today)
while stack:
    a, b = stack.pop()
    items = fetch(query, after=a, before=b)
    if len(items) >= CAP and (b - a).days > 1:
        m = a + (b - a) / 2
        stack += [(m, b), (a, m)]
        continue
    save(items)
    checkpoint(query, a, b, len(items))
```

A search returns at most $C = 100$ items. For a window $W = [a, b)$ with $N(W)$ articles:

- $N(W) < 95$ or $|W| \leq 1$ day: keep W;
- otherwise split at $m = a + \frac{b-a}{2}$.
- Depth $\leq \lceil \log_2 31 \rceil = 5$ per month; $2L - 1$ requests for L windows kept.
- 55 searches (country editions $\times$ languages).

Scoring relevance

Step 1: is it really *Central Bank Digital Currency*?

```
TERMS = {
  "CBDC": r"\bCBDCs?\b|\bwCBDCs?\b|\brCBDCs?\b",
  "central bank digital currency":
      r"central[\s-]+banks?(?:['']s?)?[\s-]+"
      r"digital[\s-]+currenc(?:y|ies)",
  "digital euro": r"digital(?:e[rn]?)?\s+euro\b"
      r"|euro\s+numérique|euro\s+digitale?\b",
  "DCEP (e-CNY)": r"\bDC/?EP\b",
  ...
}
```

- "CBDC" is also a *skin disease*; "DCEP" a chemical.
- Ambiguous terms $t \in \mathcal{A}$ count only with monetary context:

$$\text{count}(t) = n_t \cdot 1\big[t \notin \mathcal{A} \vee \text{context}\big]$$

- context = *currency, payment, central bank, monnaie, Geld...*
- *digital dollar*, *Sand Dollar*: reported, never scored.

Step 2: how central is CBDC to the text?

Each available field f gets a centrality $c_f \in [0, 1]$:

$$c_{\text{title}} = 1[\text{CBDC term in title}] \qquad c_{\text{kw}} = 0.4 \cdot 1[\text{CBDC in keywords}]$$

$$c_{\text{abs}} = \frac{\#\{\text{sentences mentioning CBDC}\}}{\#\{\text{sentences}\}} \qquad c_{\text{body}} = \tfrac{1}{2} \min\left(1, \frac{\rho}{3}\right) + \tfrac{1}{2}\, \sigma$$

ρ = mentions per 1,000 words; σ = share of the ten slices of the text that mention CBDC.

$$\boxed{\text{score} = 100\left(0.6 \max_{f \in F} c_f + 0.4 \frac{1}{|F|} \sum_{f \in F} c_f\right)} \qquad F = \text{fields available}$$

One field clearly about CBDC suffices; a passing mention stays low; missing fields are not penalised.

Step 3: labels, and how good is the filter?

Label (all texts kept)	Texts	Theme	95% CI
Main theme (≥ 50)	15,464	29/30	0.83–0.99
Secondary theme (25–49)	2,245	11/29	0.23–0.56
Mentioned only (< 25)	489	1/30	0.01–0.17
Press, text not available	13,446	2/19	0.03–0.31
Ambiguous terms only	1,709	3/10	0.11–0.60
Off-topic (homonyms, no match)	14,803	0/20	0.00–0.16
Dated before 2014 (checked: false)	32	–	–

Theme = share read as having CBDC as a theme, in a stratified sample of 138 texts read blind to the label; Wilson intervals.

From here on: main + secondary theme (17,709 texts).

Strata h weighted by size N_h; T = theme labels:

$$\widehat{P} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h \in T} N_h}, \qquad \widehat{R} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h} N_h \hat{p}_h}$$

- Precision 0.89, recall 0.89.
- Main theme is reliable; secondary theme is noisy $\Rightarrow$ robustness on main theme only.
- Most missed texts: press without text ($\approx$ 1,415).

The corpus

Sources and completeness

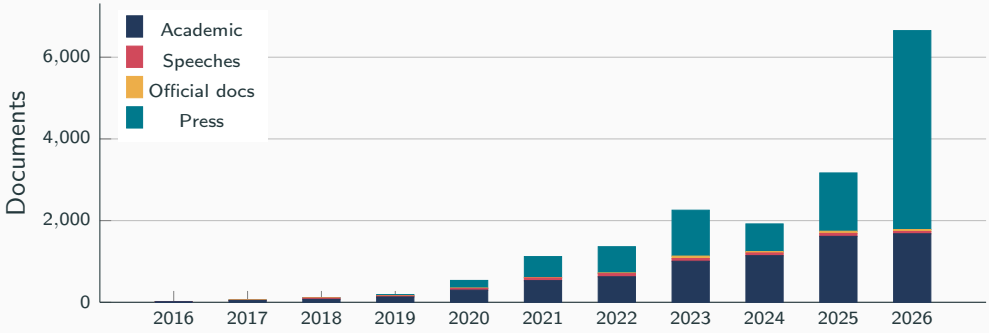
Database	Documents
Google News	9,496
OpenAlex	7,266
Semantic Scholar	4,000
DOAJ	485
BIS central bankers' speeches	465
Central banks & authorities (crawl)	239
ECB speeches	108
arXiv	90
HAL	83

Merged duplicates count in each database.

Type of text	Documents
Academic	7,463
Speeches	511
Official docs	239
Press	9,496
Total	17,709

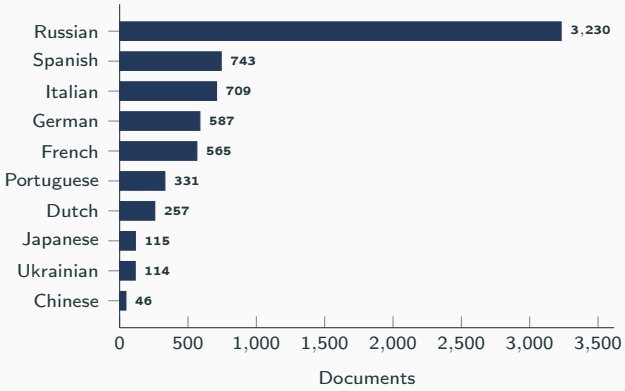
Field available	Share
Exact day of publication	86%
Issuer / outlet identified	97%
Abstract	32%
Full text	26%
Keywords	42%
Reference list	16%

A literature that took off after 2020



4 documents date from before 2016; 2026 runs to October. Press counts follow the search engine's coverage, which favours recent articles: read trends on academic texts.

Languages



- English dominates: 10,733 texts (61%); chart shows the others.
- Second language: Russian (3,230 texts, of which 2,712 press).
- Speeches kept in their original language.

Academic literature

Main outlets

SSRN Electronic Journal	876
Zenodo (CERN repository)	284
RePEc: Research Papers in Economics	236
Springer eBooks	139
Routledge eBooks	108
CyberLeninka (Russian repository)	81
Ekonomika i Upravlenie	50

Working papers (SSRN, RePEc) outweigh journals.

Bibliometric laws. Growth 58%/year (2017–2025). Lotka: $f(n) \propto n^{-\mathbf{3.0}}$, 81% of authors publish once. Bradford: 24 core sources hold **1/3** of the papers (zones 24 : 469 : 2,005).

Most cited papers (global citations)

Bordo (2017)	Central Bank Digital Currency and the Future of Monetary...	517
Auer (2020)	Rise of the Central Bank Digital Currencies: Drivers, App...	454
Agur (2019)	Designing Central Bank Digital Currencies	433
Andolfatto (2020)	Assessing the Impact of Central Bank Digital Currency on...	404
Barrdear (2016)	The Macroeconomics of Central Bank Issued Digital Currenc...	341

Who speaks: central banks, authorities, press

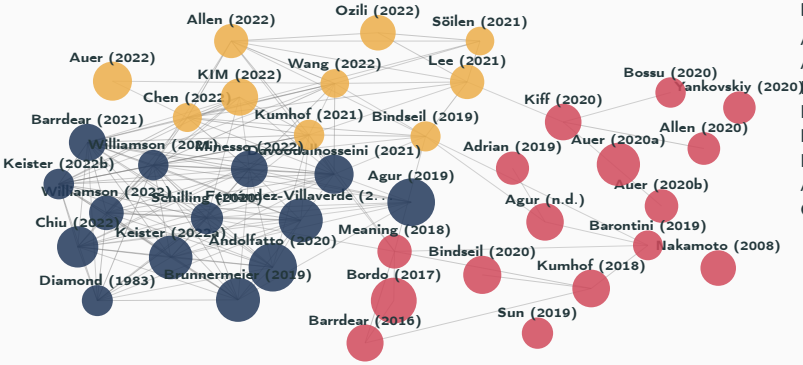
Speeches	
European Central Bank	118
Deutsche Bundesbank	63
Bank of France	49
Hong Kong Monetary Authority	23
Federal Reserve Board	21
Bank of England	21
Bank of Italy	19
Bank of Japan	14

Official documents	
European Central Bank	98
Bank of Japan	27
Sveriges Riksbank	22
Deutsche Bundesbank	21
European Commission	15
Swiss National Bank	9
European Parliament	8
The White House	7

Press outlets	
Ledger Insights	343
CoinGeek	89
Euronews.com	83
Reuters	82
TradingView	78
South China Morning Post	69
European Central Bank	65
Global Government Finance	59

Mapping the field

Co-citation: the intellectual base



Most cited within the corpus

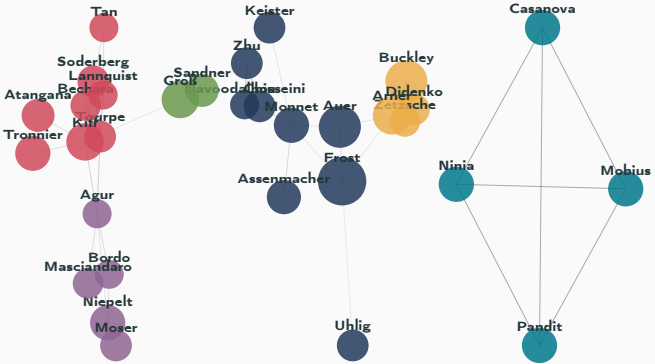
Andolfatto (2020)	359
Agur (2019)	345
Bordo (2017)	314
Brunnermeier (2019)	283
Fernández-Villaverde (2020)	283
Keister (2022)	276
Auer (2020)	272
Chiu (2022)	241

Clusters (largest members):

- Andolfatto (2020), Agur (2019), Fernández-Villaverde (2020), Brunnermeier (2019)
- Bordo (2017), Auer (2020a), Bindseil (2020), Kumhof (2018)
- Auer (2022), KIM (2022), Ozili (2022), Lee (2021)

2,749 papers with reference lists, 28,897 distinct references; 40 most co-cited shown, 3 clusters, $Q = 0.12$.

Co-authorship: research communities



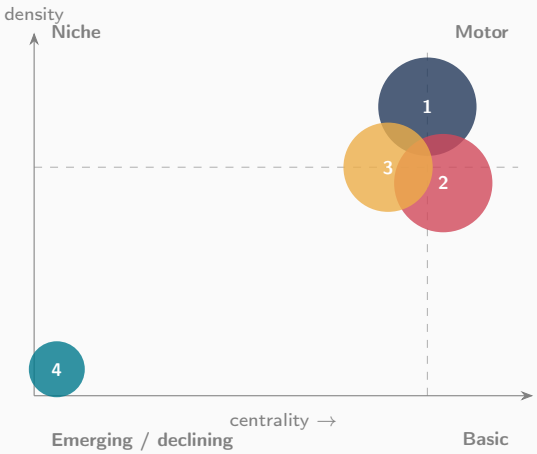
Most productive authors

Peterson K Ozili	47
Jon Frost	27
Ulrich Bindseil	23
Ross P. Buckley	19
Raphael Auer	19
Sangam Das	18
Annelieke Mooij	17
Heng Wang	15

10,733 authors; 49% of papers co-authored (mean team 2.0); 17% international. Top countries: RU 602, US 509, CN 410, IN 349, GB 343.

Papers with a central bank, BIS, IMF or World Bank author: 5% (13% in 2014–2019, 4% in 2023–2026), but 31% of citations.

Thematic map: which themes drive the field?



Keyword clusters (Callon), with $e_{ij} = c_{ij}^2/(c_i c_j)$:

$$\text{centrality} = 10 \sum_{i \in C,\, j \notin C} e_{ij}, \qquad \text{density} = \frac{100}{|C|} \sum_{i,j \in C} e_{ij}$$

	Theme (top keywords)	**Quadrant**
1	digital transformation in financial services, monetary policy, moneta...	motor
2	blockchain technology applications and security, cryptocurrency, digi...	basic
3	fintech, crowdfunding, digital finance, financial inclusion, digital...	niche
4	digital ruble, bank of russia, digital economy	emerging or declining

How the themes evolve

Keyword (% of the period's papers) *papers in period*	**2014–2019** *232*	**2020–2022** *1,360*	**2023–2026** *5,368*	**Change**
financial inclusion	5.6	9.2	15.9	+6.7
digital euro	3.0	6.8	9.7	+2.9
financial stability	15.9	10.7	12.6	+1.9
blockchain	12.1	11.7	13.3	+1.6
digital ruble	0.0	8.8	9.6	+0.8
stablecoin	1.7	10.7	11.0	+0.3
digital transformation in law	11.2	15.7	15.0	−0.7
banking stability, regulation, efficien...	14.2	10.5	9.6	−0.9
fintech, crowdfunding, digital finance	50.4	57.1	55.4	−1.7
digital payments	8.6	11.1	8.7	−2.4
monetary policy and economic impact	35.8	15.1	12.4	−2.7
payment systems	11.6	12.6	8.1	−4.5

Sorted by change between the last two periods. Darker = more frequent.

New in each period (c-TF-IDF, periods as classes): **2014–2019**: *cash*, *cryptocurrencies*, *e-krona*, *riksbank*, *bitcoin*, *legal*; **2020–2022**: *china*, *euro*, *cash*, *blockchain*, *world*, *cryptocurrencies*; **2023–2026**: *euro*, *legal*, *blockchain*, *finance*, *framework*, *inclusion*.

Text as data: what, why, how, when

Toolbox: the method chosen for each question

Question	Used	Also tested, and why not
What?	topics: BERTopic (SBERT, UMAP, HDBSCAN), labels by c-TF-IDF; co-word and Callon map; covariate logit on topic shares	LDA / STM: bag of words, one language at a time; their covariate model is applied to BERTopic instead
Who?	co-citation, co-authorship, Lotka, Bradford, affiliations; log-odds with Dirichlet prior	plain TF-IDF: no shrinkage of rare words
Why?	motivations: validated codebook, $\kappa = 0.53$; covariate logit	SBERT, NLI, local LLMs: $\kappa \leq 0.33$
How?	tone: FinBERT on CBDC sentences, $\kappa = 0.52$; stance: levels by reading, changes by a local LLM on a stratified sample, $\kappa = 0.41$	stance: LLM Qwen2.5 7B 0.41, NLI zero-shot 0.28, FinBERT tone 0.17, SBERT classifier 0.16
When?	vocabulary: c-TF-IDF by period; Kleinberg bursts; event study with placebo dates	Poisson z-test: 24% false positives
All	six robustness specifications; every coder checked on hand-coded texts	

What: topics found by meaning (SBERT + BERTopic)

	Topic (top c-TF-IDF terms)	**Texts**	**% of the period's texts**			**Adjusted, points**	
			2014–2019	2020–2022	2023–2026	change	CB
1	retail, implementation, interest, inclusion	1,495	13.0	18.0	18.4	+3.4	**-17.4**
2	euro, european, ecb, europe	1,050	19.6	13.8	12.3	+2.3	**+71.0**
3	cryptocurrencies, traditional, cash, legal	771	14.5	10.9	8.8	**-7.8**	**-8.7**
4	pound, japan, retail, pilot	636	16.3	10.8	5.9	**-11.6**	**-4.5**
5	ruble, russian, russia, legal	627	0.9	6.3	8.0	**+6.7**	**-6.9**
6	blockchain, privacy, offline, security	584	6.3	7.5	7.3	-0.1	**-7.4**
7	china, e-cny, rmb, yuan	432	0.9	6.3	5.3	**+4.2**	**-5.8**
8	cryptocurrencies, cryptocurrency, bitcoin, cr…	362	6.6	3.9	4.5	**-2.8**	**-3.3**
9	debt, dollar, reserve, crisis	342	3.3	3.8	4.4	+0.8	**-2.2**
10	india, rupee, intention, indian	324	0.0	1.7	4.9	+4.8	**-4.2**
11	fintech, services, finance, innovation	260	4.2	1.9	3.5	-0.9	-1.0
12	indonesia, rupiah, islamic, legal	205	0.6	1.7	2.9	**+2.2**	**-1.6**

8,213 texts (research, speeches, official), multilingual SBERT; 20 topics (24% outliers reassigned). Labels: $W_{t,c} = \frac{\mathbf{tf}_{t,c}}{|c|} \log\left(1 + \frac{A}{f_t}\right)$. Adjusted: logit of topic membership on period, actor, English (the covariate model of an STM); change = 2023–26 vs 2014–19; CB = central banks vs academics; **bold**: 95% CI excludes 0.

Topics tied to one jurisdiction: 41–43% of texts in every period, but the mix shifts: *pound*, *euro* fall; *ruble*, *india*, *china* rise.

Who says what: academics, central bankers, press

Distinctive words, log-odds ratio with informative Dirichlet prior (Monroe, Colaresi & Quinn, 2008), group i vs. others j:

$$\delta_w = \log \frac{y_{iw} + \alpha_w}{n_i + \alpha_0 - y_{iw} - \alpha_w} - \log \frac{y_{jw} + \alpha_w}{n_j + \alpha_0 - y_{jw} - \alpha_w}, \qquad z_w = \delta_w \Big/ \sqrt{\frac{1}{y_{iw}+\alpha_w} + \frac{1}{y_{jw}+\alpha_w}}$$

Academics (508,067 words)		**Central bankers** (232,446 words)		**Press** (23,343 words)	
research	24	payments	39	euro	50
analysis	22	area	24	rupee	45
legal	21	europe	24	yuan	43
adoption	21	european	24	pilot	39
development	20	euro	23	rbi	38
blockchain	20	inflation	21	china	34
cryptocurrencies	19	money	20	ecb	33
design	19	cash	20	launch	31

English texts; z-scores; $\alpha_w = \alpha_0 \cdot$ corpus frequency of w; topic words removed. Academics: title + abstract; central bankers: speeches; press: headlines.

Why study CBDCs? Coding stated motivations

```
MOTIVES = {
  "Financial inclusion":
      r"financial\s+inclusion|unbanked|underbanked"
      r"|access\s+to\s+(?:financial\s+services|finance)",
  "Monetary sovereignty":
      r"monetary\s+sovereignty|global\s+stablecoin"
      r"|\blibra\b|big\s*tech|dollari[sz]ation",
  "Decline of cash":
      r"declin\w*\s+(?:in|of)\s+(?:the\s+)?cash"
      r"|cashless|use\s+of\s+cash",
  ...
}
```

- Motivations are stated in the introduction (else abstract, else opening of the text).
- 10 motivations, codebook built by reading introductions.
- For text t, motivation k, group G:

$$m_k(t) = 1[\text{regex}_k \text{ matches } t]$$
$$\text{share}_k(G) = \tfrac{1}{|G|} \sum_{t \in G} m_k(t)$$

- 6,881 texts coded; 76% state at least one motivation.
- Validated on 50 held-out texts (reading-based coding, AI-assisted): precision 0.78, recall 0.46, $\kappa = 0.53$.

Why study CBDCs? Results

Motivation	% of texts 2014–2019	2020–2022	2023–2026	Adjusted effect, points [95% CI] Change 2023–26 vs 2014–19	Central banks vs academics
Technology (DLT, programmability)	31.8	32.0	35.8	+0.6 [-6, +7]	**-19.7 [-22, -17]**
Financial stability	26.0	21.5	25.3	-0.0 [-6, +5]	-0.1 [-3, +3]
Monetary policy	33.3	22.0	19.7	**-15.0 [-21, -9]**	**-7.0 [-10, -4]**
Cross-border payments	12.0	16.7	22.4	**+9.9 [+5, +14]**	**-6.3 [-9, -4]**
Privacy and surveillance	9.7	13.4	18.7	**+8.8 [+4, +12]**	**-4.8 [-7, -2]**
Payment efficiency	10.1	11.4	18.3	**+8.0 [+4, +11]**	**-3.3 [-6, -1]**
Financial inclusion	3.9	8.8	17.5	**+13.2 [+10, +15]**	**-7.5 [-10, -5]**
Monetary sovereignty	15.9	16.7	12.8	-3.1 [-8, +1]	-1.7 [-4, +1]
Geopolitics and sanctions	2.7	6.1	11.9	**+9.5 [+7, +11]**	**+3.1 [+0, +6]**
Decline of cash	12.0	7.7	6.4	**-4.7 [-9, -1]**	**+2.4 [+0, +5]**

Logit of each motivation on period, actor and English; average marginal effects, robust SEs. **Bold**: CI excludes 0.

- Rising: financial inclusion (+13 pts), cross-border payments (+10 pts), geopolitics and sanctions (+9 pts). Falling: monetary policy (−15 pts), decline of cash (−5 pts).
- Actors differ: central banks stress *Geopolitics and sanctions*, academics *Technology (DLT, programmability)*. Recall ≈ 0.46: shares are lower bounds; compare motivations and periods rather than read levels.

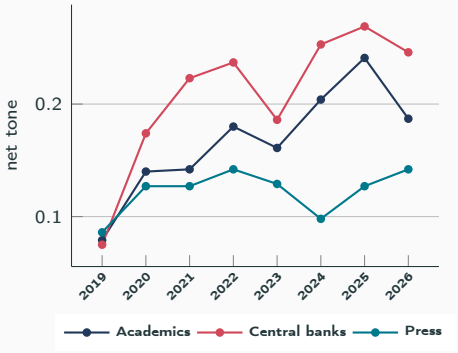
Why: is the codebook the best coder?

Coder	P	R	F_1	κ
Codebook (regex)	0.78	0.46	0.58	0.53
Codebook OR classifier	0.47	0.67	0.55	0.45
Local LLM, Qwen2.5 7B	0.38	0.58	0.46	0.33
SBERT, classifier (100 labels)	0.39	0.40	0.39	0.28
Local LLM, Qwen2.5 3B	0.35	0.28	0.31	0.20
NLI zero-shot (mDeBERTa)	0.27	0.45	0.33	0.17
SBERT, definitions (zero-shot)	0.23	0.74	0.35	0.15

50 held-out texts $\times$ 10 motivations; thresholds and classifiers set on the 100 development texts. NLI: multilingual mDeBERTa; LLMs: Qwen2.5 Instruct, 4-bit, run locally (`scripts/validate_coders.py`).

- The codebook wins: $\kappa = 0.53$; best alternative 0.33.
- Models confuse "mentions privacy" with "studies CBDCs because of privacy"; LLMs over-tag.
- Trends hold: same direction of change for 10 of 10 motivations with the high-recall coding, 10 of 10 on main-theme texts only.

How: the tone of what is said about CBDCs



	Texts	Sent.	%+	%−	Net tone
Academics	4,351	13,147	21	5	0.19 ± 0.01
Central banks	662	8,017	19	3	0.21 ± 0.01
Press	3,924	3,926	16	8	0.13 ± 0.01

FinBERT on the 25,090 English sentences that name a CBDC (central banks: speeches and official documents). Sentence tone $= P(+) - P(-)$, averaged per text, then per group ($\pm$ 95% CI).

90 sentences read (30 per predicted label): agreement 68%, $\kappa = 0.52$; positive and negative confused 2 times.

How: stance towards CBDCs

Level (read: 50 texts per actor), % [95% CI]

	n	Favourable	Critical
Academics	50	16 **[8–29]**	2 **[0–10]**
Central banks	50	52 **[39–65]**	8 **[3–19]**
Press	50	22 **[13–35]**	14 **[7–26]**

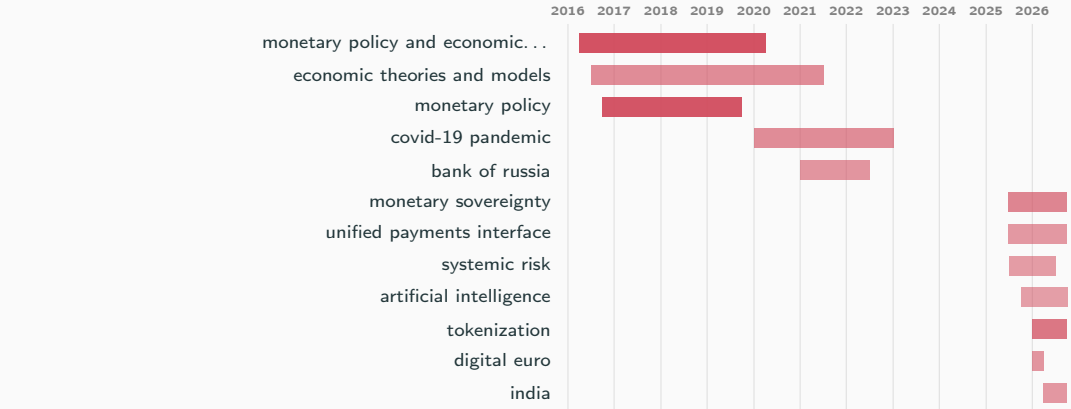
By reading: 52% of central bank texts are favourable, 16% of research; criticism is rare and mostly in the press (14%).

Change (local LLM, 829 texts sampled per actor × period), % favourable

	2014–2019	2020–2022	2023–2026	Change, pts
Academics	30	35	38	+8 **[-5, +21]**
Central banks	25	55	60	**+35 [+21, +48]**
Press	35	47	53	**+18 [+2, +34]**

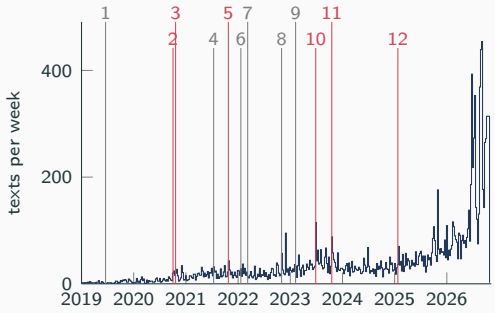
The LLM (Qwen2.5 7B, $\kappa = 0.41$, best of five coders) calls 52% of the hand-coded texts favourable against 30% by reading: its levels are inflated, so only its *changes* are used (assuming a stable bias). **Bold**: CI excludes 0.

When: which themes burst, and when



Kleinberg (2002) burst detection on keywords, by quarter: keyword w in r_t of d_t papers; normal rate $p_0 = R/D$, burst rate $p_1 = 2p_0$; cost $\sigma_i(t) = -\ln\left[\binom{d_t}{r_t} p_i^{r_t}(1-p_i)^{d_t - r_t}\right]$, entering a burst costs $\ln T$; the cheapest state path (Viterbi) gives the bursts. Darker = stronger.

When? Attention around policy events



	Date	Event	Ratio	z	p
1	Jun 2019	Facebook announces Libra	0.4	-1.3	0.77
2	Oct 2020	ECB digital euro report	**3.2**	+8.8	0.00
3	Oct 2020	Bahamas launches Sand Dollar	**2.8**	+7.7	0.02
4	Jul 2021	ECB investigation phase	1.4	+2.8	0.15
5	Oct 2021	Nigeria launches eNaira	**1.9**	+5.4	0.04
6	Jan 2022	Fed CBDC discussion paper	1.4	+2.8	0.15
7	Mar 2022	US order on digital assets	0.6	-2.6	0.94
8	Nov 2022	India e-rupee pilot	1.7	+4.6	0.07
9	Feb 2023	UK digital pound consultation	0.8	-1.3	0.77
10	Jun 2023	EU digital euro bill	**2.6**	+13.1	0.00
11	Oct 2023	ECB preparation phase	**1.9**	+8.7	0.01
12	Jan 2025	US order banning a US CBDC	**1.8**	+6.3	0.02

Baseline λ = daily rate over $[-90, -15]$ days; $E = 15\lambda$; ratio $= N_{[0,14]}/E$; $z = (N - E)/\sqrt{E}$. Placebo: on 500 random dates, 24% have $z > 1.96$ (overdispersion, trend), so p = share of placebo dates with a larger z (95th percentile: $z = 5.0$).

Robustness

Specification	Texts	Why: change, points inclusion	Why: change, points monetary policy	How: tone CB − press	How: stance fav. CB − acad.	When events
Baseline	17,709	+13.2 [+10, +15]	−15.0 [-21, -9]	+0.08	+14	6/12
Main theme only	15,464	+15.2 [+11, +18]	−11.8 [-19, -5]	+0.06	+13	7/12
English only	10,733	+14.3 [+10, +17]	−15.9 [-22, -9]	+0.08	+15	7/12
Journal articles only (research)	14,270	+15.0 [+11, +17]	−14.6 [-22, -7]	+0.08	+9	7/12
Without 2026 (partial year)	10,742	+12.9 [+9, +15]	−14.9 [-21, -9]	+0.09	+13	6/12
High-recall motivation coding	17,709	+16.2 [+12, +19]	−9.9 [-16, -4]	+0.08	+14	6/12

Changes: 2023–26 vs 2014–19, adjusted (logit, 95% CI). Tone: mean net tone gap. Stance: raw LLM labels, points. Events: $p < 0.05$ against placebo dates.

Signs and orders of magnitude hold in every specification.

Conclusion

Takeaways and limits

First results

- What: 17,709 texts; output grows 58% a year; topics follow national projects.
- Who: a core of co-cited papers (Andolfatto (2020), Agur (2019), Bordo (2017)); central bank economists write 5% of papers but get 31% of citations; European Central Bank the most vocal central bank.
- Why: rising financial inclusion (+13 pts), cross-border payments (+10 pts), geopolitics and sanctions (+9 pts); falling monetary policy (−15 pts), decline of cash (−5 pts).
- How: central banks most positive in tone and stance, and increasingly so; criticism is rare, mostly in the press.
- When: bursts move from monetary policy to sovereignty and tokenisation; 6 of 12 milestones move attention.

Limits

- Press: headlines only; coverage favours recent years; harvest cut short by rate limits.
- "Secondary theme" is noisy: results re-run on main theme only.
- Motivations: recall ≈ 0.5 (lower bounds). Tone: English only. Stance: LLM for changes only.
- One coder (AI-assisted) for all validation samples.
- Keywords partly machine-assigned; authors matched on spelling.

From this talk to the paper

Structure

	Section	Status
1	Introduction, research question	to write
2	Related work (77 reviews)	material
3	Data: corpus, relevance filter	validated
4	Methods: maps, topics, coders, tone, stance, events	done
5	Results: what, who, why, how, when	done
6	Robustness (6 specifications)	done
7	Discussion, limits	to write

Working title. *Why central bank digital currencies? Researchers, central bankers and the press, 2014–2026.*

Before submission

- Finish the press harvest and full-text downloads.
- Second human coder: the 200-item sheet is ready (`second_coder_sheet.csv`); report inter-coder κ.
- Release the dataset and code with a DOI.
- Methods are complete; what is left is data collection and a second reader, not new tools.

Candidate outlets: *Scientometrics*; *Technological Forecasting and Social Change*; *Research in International Business and Finance* (published the closest review, Hoang et al. 2023).

Backup: validation of the motivation coding

Codebook tuned on 100 texts (development set), then evaluated on 50 held-out texts; unit = first 1,500 characters of the introduction / abstract.

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}, \quad \kappa = \frac{p_o - p_e}{1 - p_e}$$

Motivation	**Read**	**Coded**	**Precision**	**Recall**	κ
Privacy and surveillance	7	6	1.00	0.86	0.91
Technology (DLT, programmability)	13	12	0.83	0.77	0.73
Monetary policy	8	7	0.71	0.62	0.61
Financial inclusion	8	3	1.00	0.38	0.50
Cross-border payments	5	3	0.67	0.40	0.46
Financial stability	5	3	0.67	0.40	0.46
Geopolitics and sanctions	9	3	1.00	0.33	0.45
Monetary sovereignty	7	2	1.00	0.29	0.41
Decline of cash	3	2	0.50	0.33	0.37
Payment efficiency	13	5	0.40	0.15	0.09
All	78	46	0.78	0.46	0.53

Read = texts where the reading found the motivation; Coded = texts flagged by the codebook. Files: `presentation/validation_motivations_dev.csv`, `_test.csv`.

Backup: how the science maps are built

- **Networks.** Nodes = references, authors or keywords; c_{ij} = documents where i and j appear together, c_i = occurrences of i. Edges weighted by association strength:

$$s_{ij} = \frac{c_{ij}}{c_i \, c_j}$$

- **Clusters.** Louvain algorithm, maximising modularity

$$Q = \frac{1}{2m} \sum_{i,j} \left[s_{ij} - \frac{k_i k_j}{2m} \right] \delta(g_i, g_j)$$

- **Layout.** Force-directed: linked nodes end up close. Size $\propto \sqrt{c_i}$, colour = cluster.

Backup: how the score behaves

Example	Score	Label
Paper, CBDC in title	87	main theme
Paper, generic title, CBDC abstract	80	main theme
Paper, one passing mention	11	mentioned only
Speech, one paragraph on it	18	mentioned only
Speech devoted to it	80	main theme
Medical case report	0	off-topic

Backup: academic sources, cursor pagination under a rate limit

```
def iter_phrase(phrase):
    cursor = "*"
    while cursor:
        d = get(f'{API}?filter=title_and_abstract'
                f'.search:"{phrase}"&per-page=200'
                f'&cursor={cursor}')
        yield from d["results"]
        cursor = d["meta"].get("next_cursor")

class Throttle:
    def wait(self):
        dt = self.interval - (now() - self.last)
        if dt > 0: sleep(dt)
        self.last = now()
```

- Each phrase searched in titles and abstracts; results merged on the source id.
- Requests spaced by $t_{k+1} - t_k \geq \Delta$ ($\Delta = 0.12\,\text{s}$ OpenAlex, $1.2\,\text{s}$ Google News, $10\,\text{s}$ GDELT).
- On HTTP 429: back-off $w_j = \min(60j,\ 900)\,\text{s}$ after the j-th consecutive failure.
- Reference lists resolved into citations (for co-citation).

Backup: full text and de-duplication

Full text

- PDFs (PyMuPDF) and web pages (trafilatura).
- Split into introduction / conclusion / section headings.
- Page metadata (date, author, outlet) fills gaps.

```
def key(r):
    if r.source_type == "academic":
        if r.doi: return "doi:" + r.doi.lower()
        return f"t:{norm(r.title)[:120]}:{r.year}"
    if r.source_type == "speech":
        return f"s:{r.date}:{norm(r.title)[:50]}"
    return "u:" + r.url
```

- Second pass: same title + year across DOIs.
- Merge: most precise date, longest abstract, union of keywords.

Backup: validation of the tone measure

30 sentences sampled per FinBERT label, read blind to the label.

FinBERT ↓ / reading →	**negative**	**neutral**	**positive**	**Precision**
negative	18	11	1	0.60
neutral	2	26	2	0.87
positive	1	12	17	0.57

90 sentences read (30 per predicted label): agreement 68%, $\kappa = 0.52$; positive and negative confused 2 times.

File: `presentation/validation_tone.csv`.

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